Backtracking

Optimization Problems

- Brute force search
  - Backtracking
- Branch and Bound
- Dynamic programming
- Divide and Conquer
- Greedy

Brute force search

Systematically enumerate all possible objects which can potentially satisfy the constraints of the problem.

- For each object: check if constraints are satisfied, and maintain the cost of the object.
- Output the optimized value.

Is the procedure correct?

How do we generate such objects?

Generating objects

- Treat objects as a collection of items/vertices/elements and consider each possible way of generating each of items/vertices/elements.

- To start with: “empty object” (does not contain any of the items/vertices/elements)
  - Fill 1st slot
  - Fill 2nd slot
  - …

Generating Objects

Find all subsets of a given set A={v1, v2, …, vn}

Idea: To explore all the choices one by one recursively. For each element, there are two options: either to include it in the subset or exclude it.

Generating Objects

What does leaves correspond to?

Find all subsets of a given set A={v1, v2, …, vn}



What does the root node say? (n=3)

Idea: To explore all the choices one by one recursively. For each element, there are two options: either to include it in the subset or exclude it.

Generating Objects

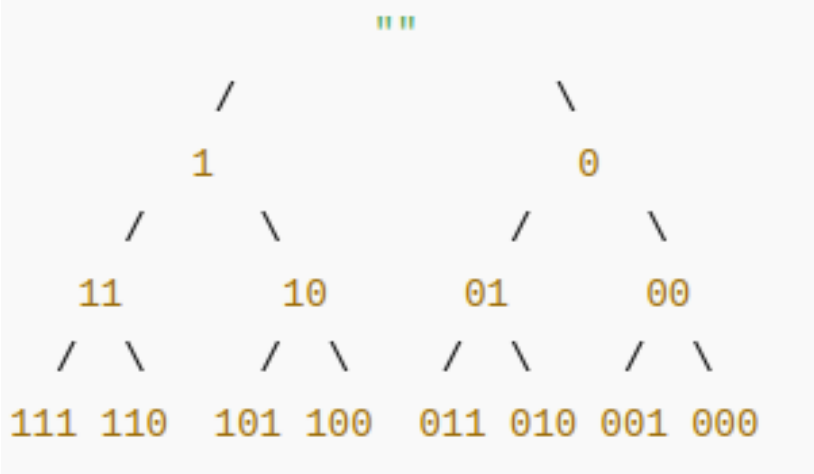
Find all subsets of a given set A={v1, v2, …, vn}



What does the root node say? (n=3)
No decisions made yet.
We are about to make a decision about v_1.

Generating Objects

Find all subsets of a given set A={v1, v2, …, vn}

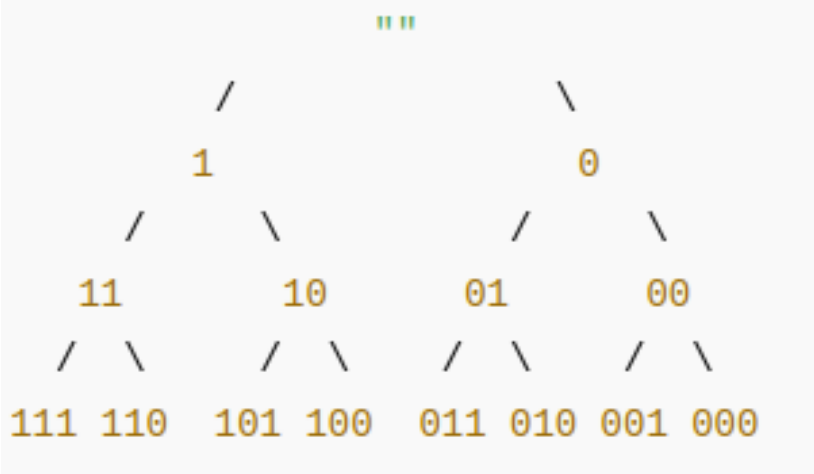


What does each level correspond to ? (n=3)

Generating Objects

State Space Tree

Find all subsets of a given set A={v1, v2, …, vn}



What does each level correspond to ? (n=3)

each level corresponds to making a decision about one element of the array.

Generating Objects

State Space Tree

Find all subsets of a given set A={v1, v2, …, vn}



What does each level correspond to ? (n=3)
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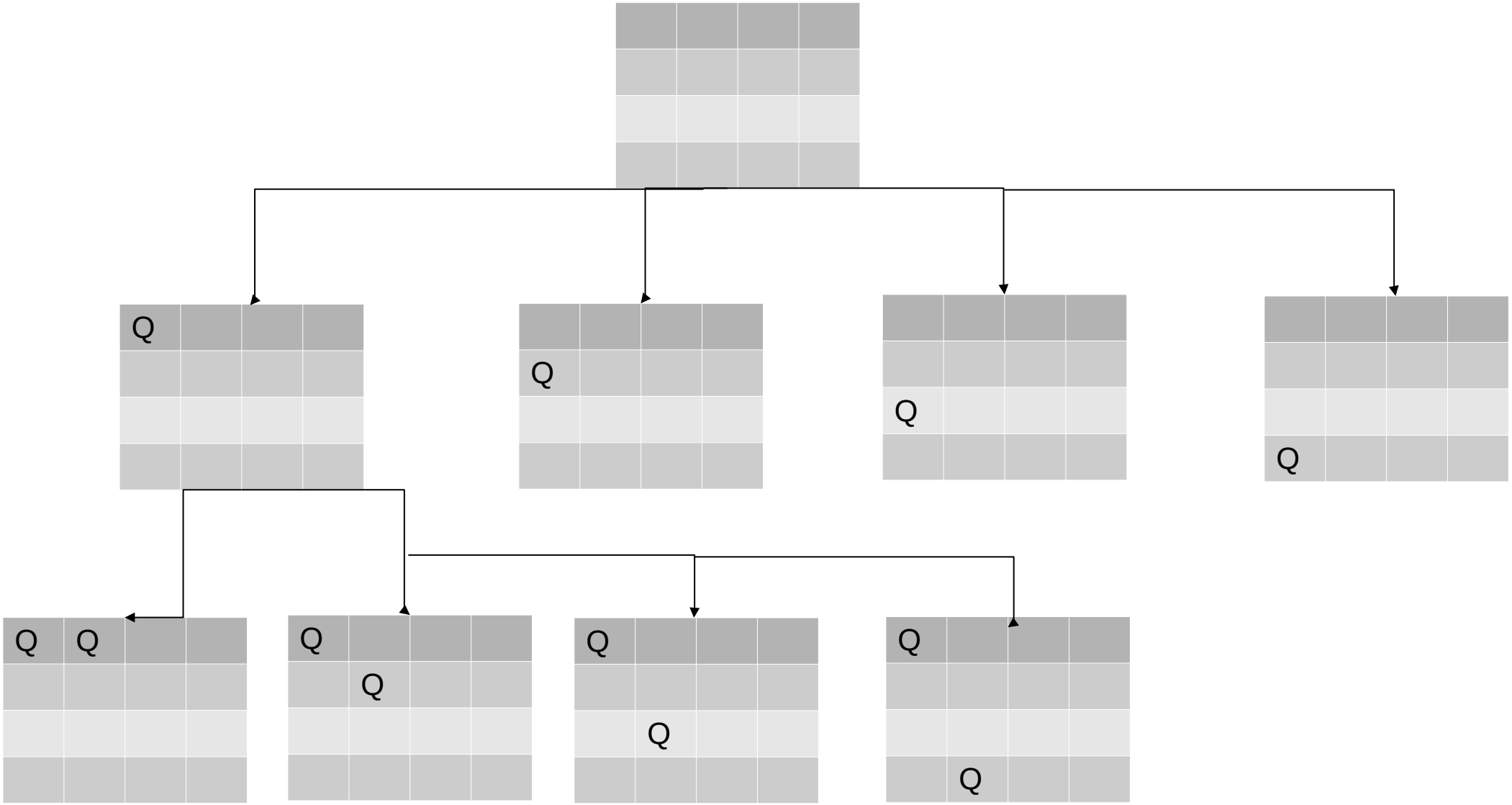
A **state space tree** is a tree that represents all possible states (partial and complete) of a search problem.

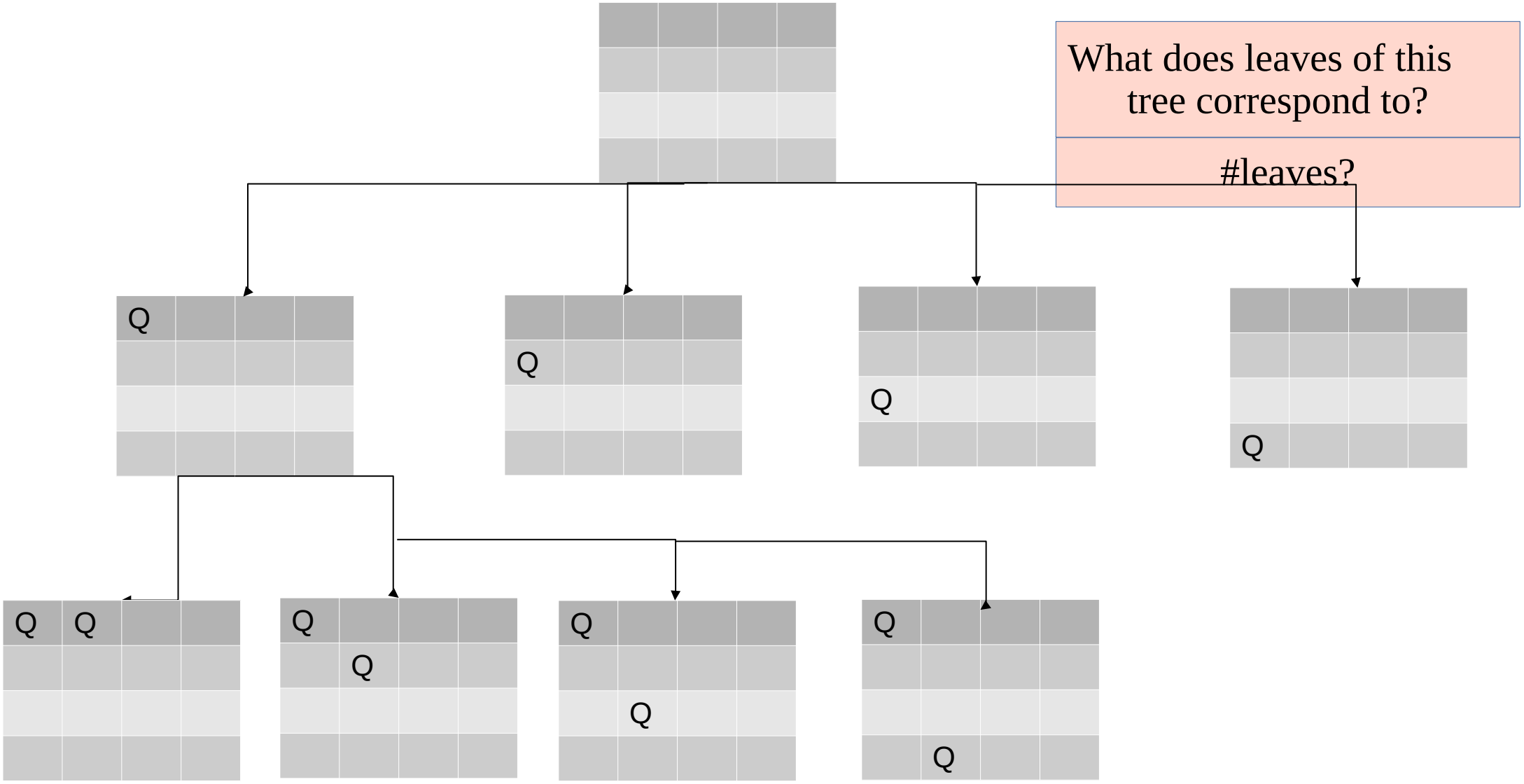
8-Queens Problem

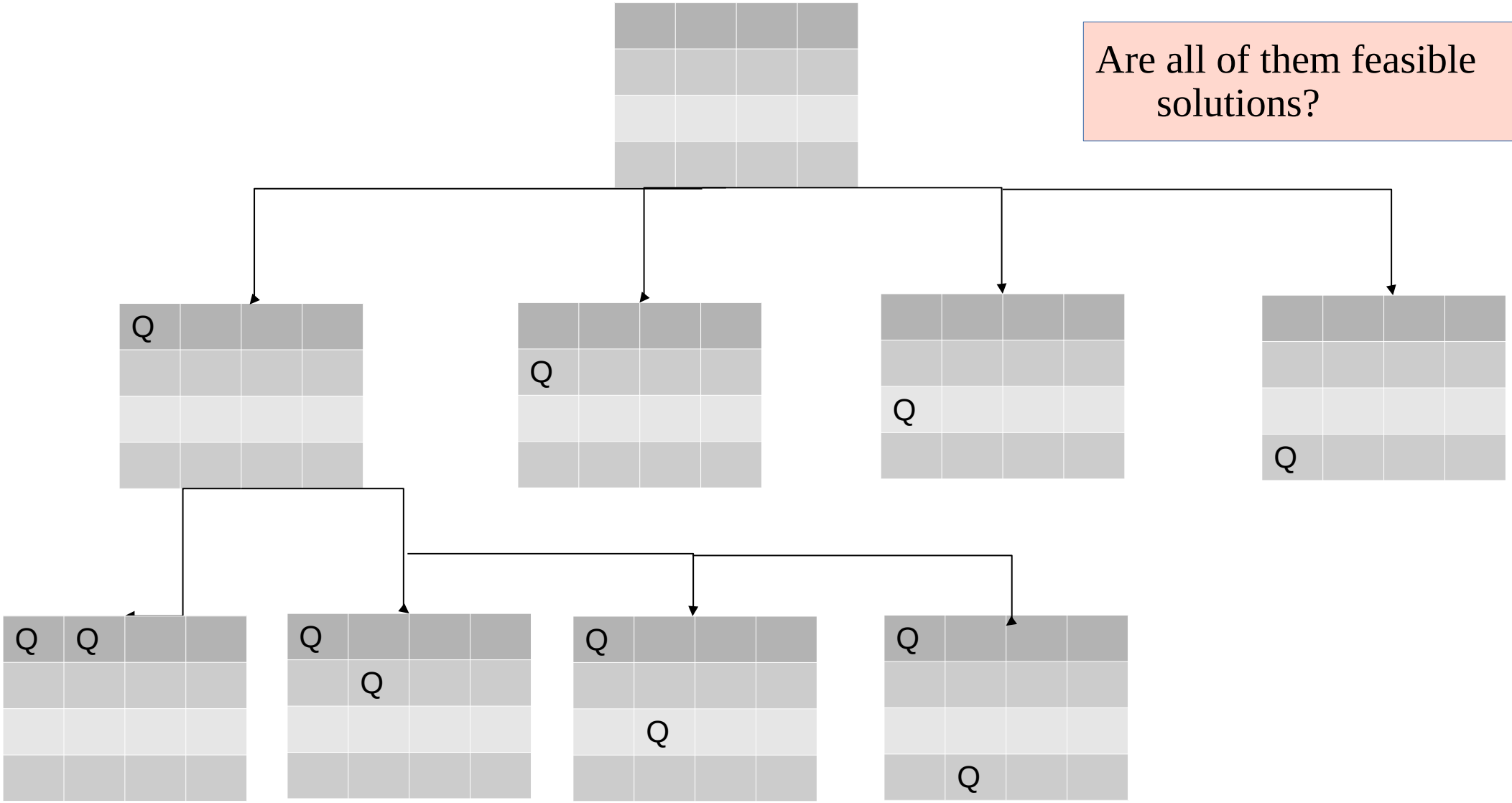
Input: An 8x8 chessboard

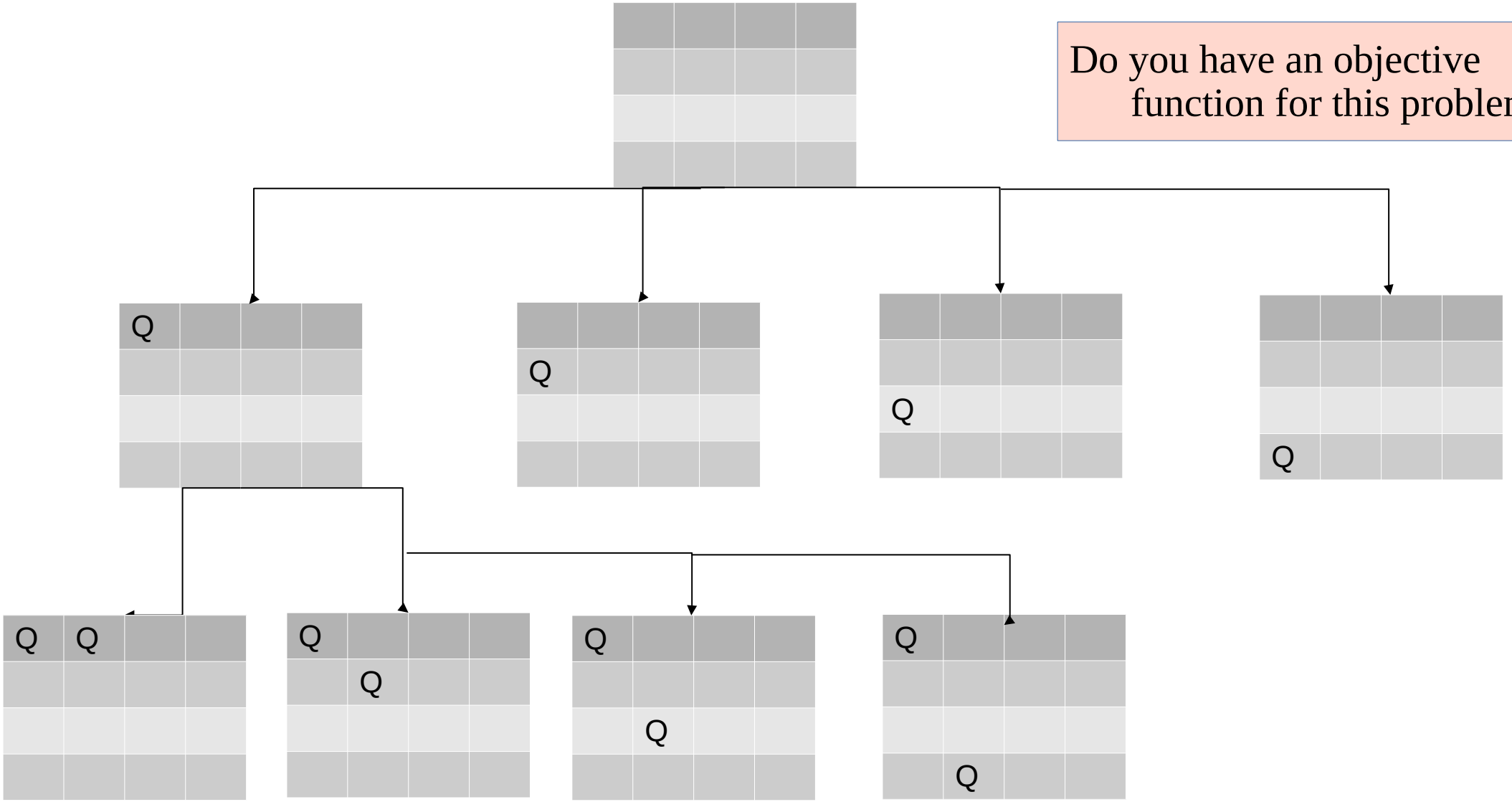
Output: Place 8 queens on the board such that no two queens are in the same row, column or diagonal.

How does the state space tree look like?


Q
Q
Q
Q
Q Q
Q
Q
Q
Q
Q


What does leaves of this tree correspond to?
#leaves?


Are all of them feasible solutions?
Q
Q
Q
Q
Q Q
Q
Q
Q
Q
Q



Do you have an objective function for this problem?

Knapsack Problem

Input: A set of n items,

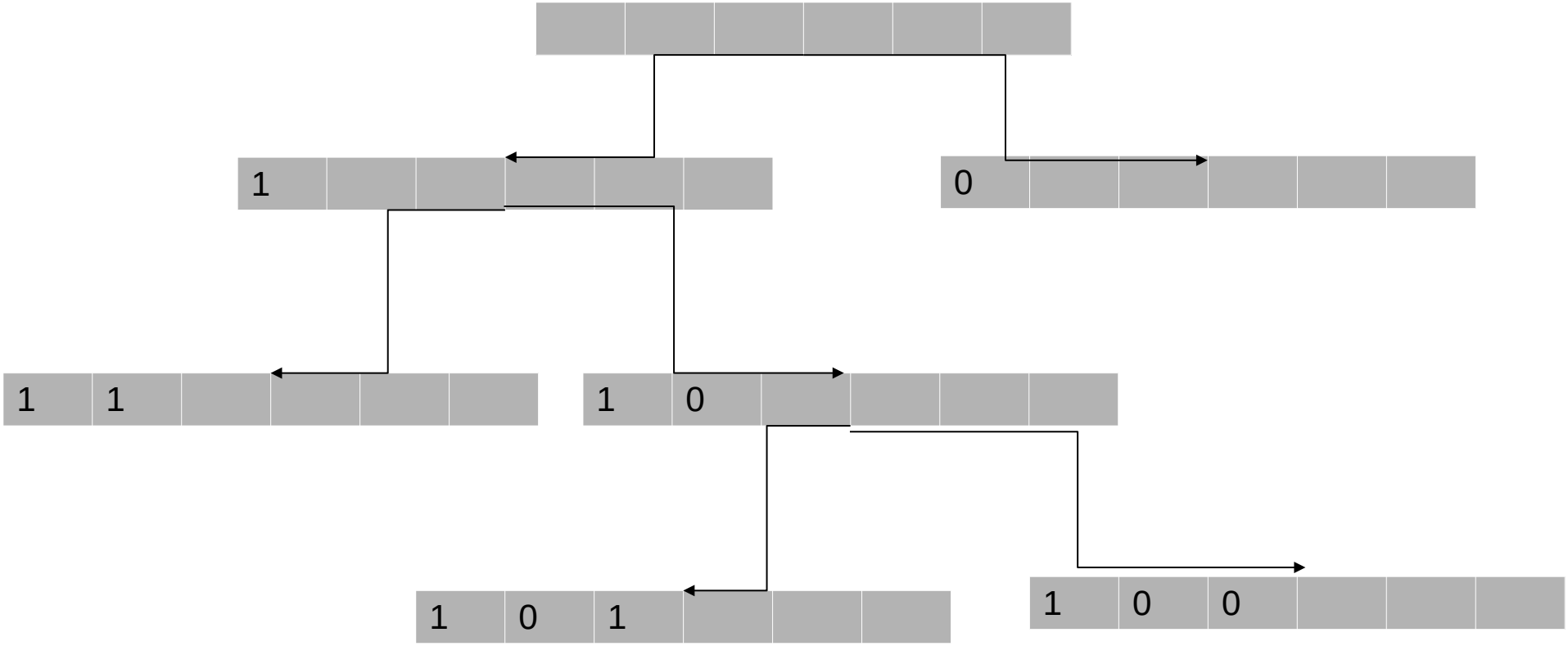
v1, v2, …, vn value of each of the items

W1, W2, …, Wn weights of each of the items

Capacity C

Output: Find a subset of items such that the total value of the selected items is as large as possible while total weight to be at most C.

How does the state space tree look like?



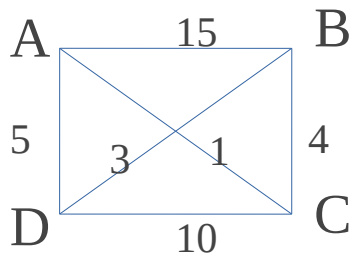
#leaves? 2^n

Travelling Salesperson Problem

Input: A weighted graph.

Output: A “tour” through the vertices of the graph that visits every vertex exactly once.

Objective function: minimize the cost of the tour in terms of distance traveled)



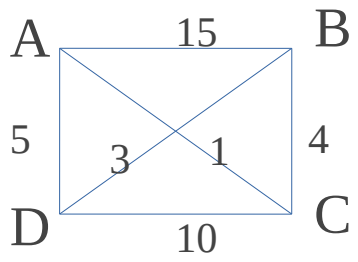
How to represent objects?

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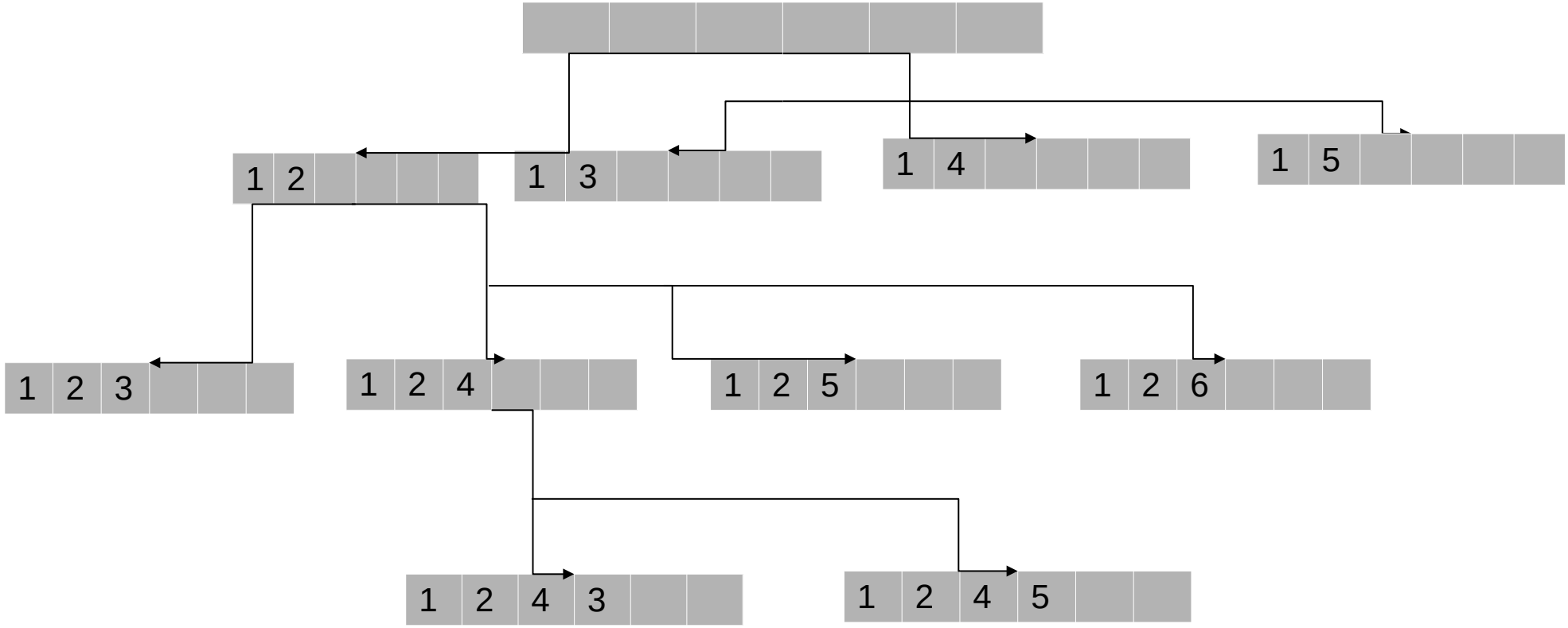
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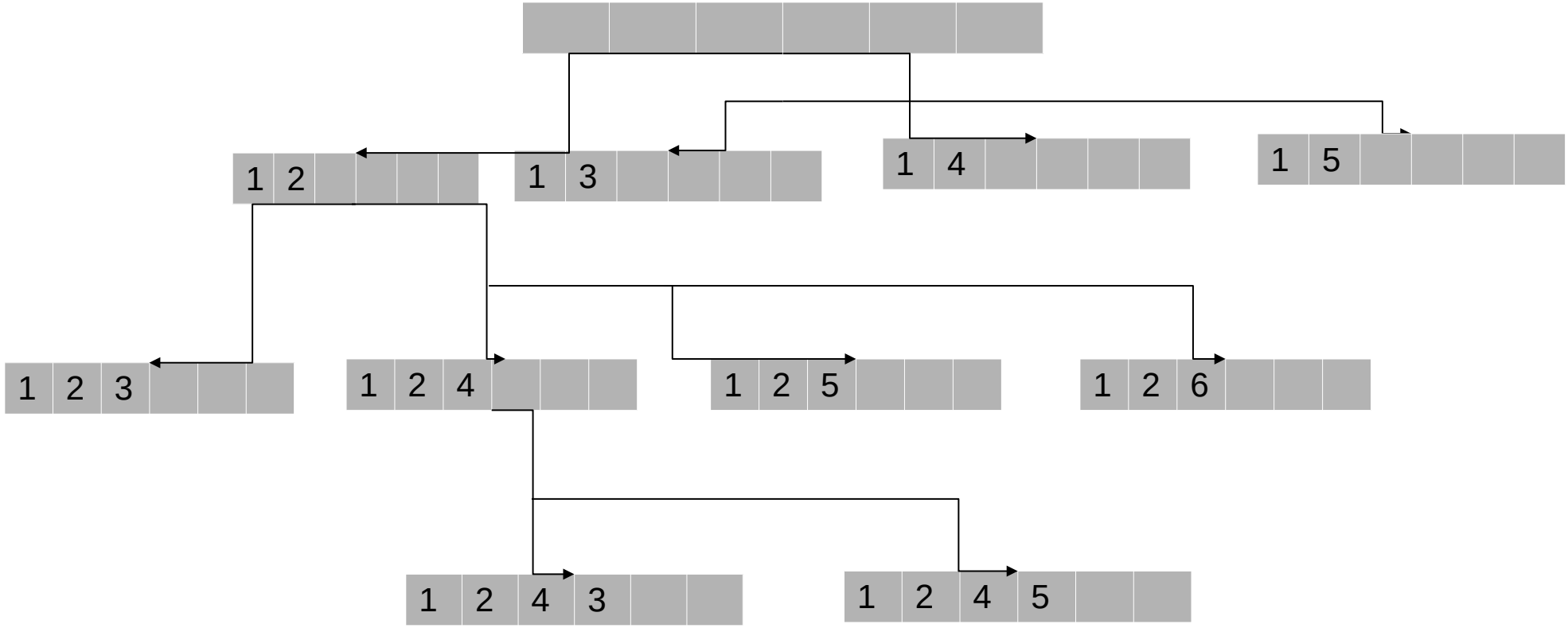


A-B-C-D-A: 34
A-C-B-D-A: 13

How does the state space tree look like?



#leaves?



#leaves? (n-1)!

Implementation

1. Representation of objects.
2. Procedure for filling in the next level (or next empty slot) of the object in all possible ways.
3. Recurse
4. For the leaf objects, we need to check if the object satisfies the constraint of the problem.
5. Further, a procedure for evaluating the cost of the object.
6. Procedure for removing the last value filled in the slot.

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- Adopts a DFS (depth-first search) strategy over the **search space**

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  Violating objects can be pruned by a local check (need not wait till we need the leaf).

- Running time?

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- Improvement?

  Violating objects can be pruned by a local check (need not wait till we need the leaf).

- Running time?

  At least the number of leaves.